MIL-B-16392C(SHIPS)
15 March 1963
SUPERSEDING
MIL-B-16392B(NAVY)
7 March 1957

MILITARY SPECIFICATION

BRAKES, MAGNET, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers class HI shockproof electric brakes of the shoe or disk type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shockproof Equipment, Class HI (High Impact), Shipboard Application, Tests for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-I -17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (For Shipboard Applications).
MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper Oxide, and Magnesium - Copper Sulfide Types, Naval Shipboard Use.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION STANDARDS
PUBLICATION ICI - Industrial Control

(Copies of this document may be obtained from the National Electrical Manufacturer's Association, 155 East 44th Street, New York 17, New York).

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification.- Brakes furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

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3.2 General requirements.- Brakes shall conform to MIL-E-917 and National Electrical Manufacturer's Association (NEMA) standards for Industrial Control. In the event of conflict, the following order of precedence shall prevail:

(a) The requirements of this specification.
(b) The applicable requirements of Government specifications referenced herein.
(c) NEMA standards

3.2.1 Insulation.- Insulation shall be class B, except that coils may be class A, B, or H.

3.2.2 Painting.- All parts not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials of the types specified in MIL-E-917 shall be painted in accordance with MIL-E-917. Touching up will be permitted for marks or scratches due to factory handling.

3.2.3 Materials.- Materials shall be first class in every respect.

3.2.3.1 Bearings and pivot pins.- Bearings and pivot pins shall be of bronze, corrosion-resisting metal, oil-impregnated insert material, or other similar materials. Nonaustenitic corrosion-resisting steel may be used for hardened pins. When lubrication is required, means shall provided without the necessity of removing the enclosure.

3.2.3.2 Material restrictions.- Brake wheels may be of cast iron. Nodular graphitic iron in accordance with MIL-I-17166 may be used for housings, pressure plates, end bells, and other structural parts where suitable.

3.3 Design.-

3.3.1 Operation.- Brakes shall be either the shoe or disc type, and shall be shunt solenoid operated. Brakes shall be spring set and electrically released.

3.3.2 Torque.- The torque rating of the brake shall be as specified (see 6.1).

3.3.3 Voltage.- Brakes shall be designed for operation at the following voltages as specified (see 6.1):

Alternating current (a.c.):
110 volts
220 volts
440 volts
Special, all others
Direct current (d.c.):
115 volts
230 volts
250 volts (175 to 355)
Special, all others

3.3.3.1 Voltage limitation.- All brakes shall be designed to operate between 90 and 110 percent of rated voltage, at rated frequency, and at rated ambient temperature.

3.3.4 Frequency.- All a.c. brakes shall be designed for operating at a frequency of 60 cycles ± 5 percent.

3.3.5 Duty.- The duty rating of the brake shall be as specified (see 6.1). A continuous-duty brake shall be used for an intermittent varying-duty motor.

3.3.5.1 Continuous duty.- Continuous-duty brakes shall be so designed as to be suitable for continuous operation without exceeding the temperature limitations specified herein.

3.3.5.2 One-hour duty.- One-hour duty brakes shall be so designed as to be suitable for operation for 1 hour without exceeding the temperature limitations specified herein by more than 20°C.

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3.3.6 Manual release feature.- All brakes shall be furnished with a manually operated release mechanism. The release mechanism shall be externally operable without removal of plates or screws. It shall be impossible to latch the brake in the released position, thus making it necessary for the operator to hold the brake in this position manually.

3.3.7 Mounting.- Brakes shall be motor mounted or bedplate mounted, as specified (see 6.1). The attachment of the brake shall not unduly complicate the removal of the rotor of the motor.

3.3.8 Adjustment.- The design shall be such that adjustments, to compensate for lining wear or to increase or decrease torque, may be made without the use of special tools. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

3.3.9 Ambient temperature.- Brakes shall be designed for a 50° centigrade (C) ambient.

3.3.10 Temperature rise.- Temperature rises shall not exceed the values specified in table I, as adjusted in accordance with 3.3.10.1 through 3.3.10.5.

Table I - Maximum temperature rise.

Name of part	Temperature rise (50°C ambient)
Shunt coils (a.c.):	
Class A insulation	60
Class B insulation	85
Class H insulation	150
Shunt coils (d.c.):	
Class A insulation	55
Class B insulation	80
Class H insulation	145
Bolted connections and terminal connections:	
Not silver plated	50
Silver plated	65

3.3.10.1 Hot-spot temperatures.- Table I is based on limiting hot-spot temperatures of 110°C for class A, 135°C for class B, and 200°C for class H insulation.

3.3.10.2 Method of measurement.- Shunt coil rises shall be measured by the resistance method. All other rises shall be measured by thermocouple or thermometer.

3.3.10.3 Temperature gradient.- If cotton or glass served wire or layer insulation is used, the allowable temperature rise for coils is 5°C. less than shown in table I.

3.3.10.4 Enclosure differential.- Coils in watertight, submersible, or explosion-proof enclosures are permitted an additional 5°C. rise over the values specified in table I.

3.3.10.5 Submarine applications.- Coils for 250-volt brakes are permitted temperature rises of 10°C. higher than specified in table I, as modified by 3.3.10.3 and 3.3.10.4.

3.3.11 Parts.-

3.3.11.1 Magnet coils.- Magnet coils for disc brakes shall be encapsulated and are preferred for shoe brakes. Terminal lugs, with binding screws, are preferable to wire leads for encapsulated coils. The minimum wall thickness of encapsulating material shall be 1/32 inch. If the coil is wound on a core tube before encapsulation, the wall thickness may be reduced to 1/64 inch in the core. The materials and methods of construction shall be as approved by the bureau or agency concerned. Coils, where of the removable type, shall be readily removable from the pole or yoke without injury to the coil. The coil shall be rigidly held so as to prevent damage to the coil leads. Magnet coil shall be marked for ready identification.

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3.3.11.2 Resistors.- Resistors shall conform to MIL-R-15109. Resistors shall be of the unit type, so constructed and installed that each unit can be readily replaced. Resistors shall be mounted within the brake or control enclosure. Provision shall be made for readily connecting the leads to the resistor steps and for preventing the leads from coming in contact with the heated resistor.

3.3.11.3 Springs.- Springs shall be designed to insure the proper functioning of the equipment to which attached. Springs shall not be stressed beyond their fatigue limits.

3.3.11.4 Rectifiers.- Rectifiers shall conform to MIL-R-18281.

3.3.11.5 Terminal leads.- All terminal leads shall be made of flexible, stranded, insulated wire. An insulated sleeve or other suitable means shall be provided to prevent abrasion of the lead insulation by metallic edges of the frame, terminal box, or cable clamp.

3.3.12 Operational requirements.-

3.3.12.1 Inclination.- Brakes shall be designed to operate when inclined in any direction at an angle of 45 degrees from the normal position.

3.3.12.2 Vibration.- Equipment shall be designed to withstand type I vibration tests of MIL-STD-167 without mechanical damage or malfunctioning.

3.3.12.3 Shockproofness.- Brakes shall be capable of withstanding class HI shock in accordance with MIL-S-901. Brakes shall be designed to perform satisfactorily before, during, and after the periods of shock. Such performance shall not depend upon mechanical shock mounts or the use of a device or devices by which the operator can lock the brake in position.

3.3.13 Insulation resistance.- Insulation resistance shall not be less than 10 megohms.

3.3.14 Dielectric strength.- Equipment shall be capable of withstanding dielectric tests as specified in NEMA Publication ICI-2.42. D.C. shunt coils not protected by a shunt resistor or other equally effective means shall be capable of withstanding, for a period of not less than 1 second, a dielectric test voltage of 10 times rated circuit voltage plus 1000 volts. Rated voltage for submarine applications is the nominal voltage.

3.4 Enclosures.- Enclosures shall be of the following types, as specified (see 6.1):

(a) Open
(b) Dripproof
(c) Watertight
(d) Submersible (15 ft-30 minutes)
(e) Explosionproof (Class I, Group D)

3.4.1 Detail requirements.- Enclosures shall conform to NEMA Publication ICI for the designated type and the following:

3.4.1.1 Dripproof.- Dripproof enclosures shall be designed to meet NEMA requirements when inclined to any angle not greater than 45 degrees from the normal position. These requirements shall apply to the electrical portion of brake only. The mechanical portion needs only to provide against injury to personnel or damage from foreign objects.

3.4.1.2 Moisture drain.- Watertight and submersible enclosures shall have a moisture drain in the bottom. The opening shall be closed with a 1/4-inch or larger pipe plug.

3.4.1.3 Identification plates.- Each brake shall be provided with an identification plate mounted on an exterior part not renewed during its life. Plates shall be brass, corrosion-resistant steel, or anodized aluminum of not less than 0.030 inch in thickness and shall be marked by etching, engraving, stamping, or photoprocess. Etching, engraving, or stamping shall be not less than 0.003 inch deep. The minimum data marked on identification plates for brakes shall include the following:

(a) Manufacturer's name and catalog number.
(b) Voltage, frequency, and torque rating.

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(c) Blank space for Government inspector's official stamping.
(d) Federal stock number.

3.5 Repair parts.-

3.5.1 Interchangeability.- Each repair part shall be equal to the original and suitable for replacement without special tools, and except for brake linings, without the removal and replacement of rivets or pins dependent upon a force fit.

3.5.2 Quantity of repair parts.- Repair parts shall be furnished in sets as shown in table II (number of sets shall be as specified (see 6.1)). For one to five identical brakes per ship, one set of repair parts shall be furnished. For six to twenty identical brakes per ship two sets of repair parts shall be furnished. For more than twenty identical brakes per ship three sets of repair parts shall be furnished.

Table II - One set of repair parts

Part
Complete replacement of all springs
Complete replacement of friction linings
Complete replacement of rivets plus 10 percent
Shunt coils, one of each size and type
Rectifiers, one of each size and type

3.6 Drawings.- Drawings shall conform to MIL-D-963. The approval activity for master drawings shall be the Bureau of Ships. Certification data may be approved by the ordering activity.

3.6.1 Master drawings.- Master drawings shall conform to class A of MIL-D-963, and shall contain the following minimum data:

(a) Outline, not necessarily to scale, of the side and end views of the brake giving dimensions and essential details of the enclosure. (Note: In the interest of clarity, views in addition to those specified herein may be shown.)
(b) Outline, not necessarily to scale, of the side view of the brake showing the outline of the equipment within the enclosure and referencing the parts to the piece numbers of the List of Material. (Note: Outlines of the equipment within the enclosure may be incorporated with the outlines of the enclosure provided that clarity of detail is not lost.)
(c) List of Material which shall include the enclosure and all components and parts, manufacturer's identification numbers, and applicable specifications.
(d) Description of adjustments which can be made on the brake, such as torque adjustment, adjustment for lining wear, and replacing linings.
(e) A list of repair parts (see table II) with piece numbers for each item, the name of each item, the quantity of each item recommended for one set of repair parts, and the manufacturer's ordering identification for the item.
(f) Table of applicable shunt coil catalog numbers with corresponding voltage, sealed watts, ohms, maximum pickup, and sealed amperes for a.c. coils and rating of series resistor, if used.
(g) Description of operation of the brake including any necessary views of working parts.
(h) Weight of the brake.
(i) List of exceptions to this specification.
(j) Descriptive data of the brake, which shall include the torque rating, voltage, phase, duty, ambient temperature, and class of insulation.

3.6.2 Certification data.- Certification data shall conform to class B of MIL-D-963, and shall contain the following minimum data:

(a) Navy contract or shipbuilder's order and item number.
(b) Brake manufacturer's order number.
(c) Brake master drawing number.
(d) Brake catalog numbers.
(e) Applicable specification.
(f) Voltage and torque rating.
(g) Number and identification of ships involved.

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(h) Application.
(i) Quantity per contract and per ship.
(j) List of repair parts.
(k) Title block.

3.7 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Tabulation of inspection.- Qualification tests, comparison inspection and quality conformance inspection shall be as shown in table III.

Table III - Inspection agenda.

Description and order of test	Qualification inspection	Comparison inspection	Quality conformance inspection	Applicable requirement	Paragraph test
Examination	X	X	X	3.2.3, 3.7	4.6
Weight	X	X			4.7.1
Creepage and Clearance	X				4.7.2
Resistance	X	X	X		4.7.3
Power input	X	X	X		4.7.4
General Operation	X	X	X		4.7.5
Inclination	X	X		3.3.12.1	4.7.6
Voltage variation	X	X		3.3.3.1	4.7.7
Temperature rise	X	X		3.3.10	4.7.8
Vibration	X	-		3.3.12.2	4.7.9
Insulation resistance	X	X		3.3.13	4.7.10
Dielectric	X	X	X	3.3.14	4.7.11
Shock	X			3.3.12.3	4.7.12
Torque	X	X			4.7.13
Explosionproof (enclosures)	X	-		3.4	4.7.14
Effectiveness of enclosure	X	X		3.4	4.7.14.3

4.3 Qualification tests.-[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in table III.

4.4 Comparison inspection.- One year after qualification inspection and each year in which the brakes are offered for delivery thereafter, one brake of each design shall be subjected to the comparison inspection shown in table III. These tests are not required to be conducted on more than one out of fifty brakes of a given design manufactured over a period of three years. Three copies of the test report shall be forwarded to the Bureau of Ships, Department of the Navy, Washington 25, D.C.

4.5 Quality conformance inspection.- Each brake shall be subjected to the quality conformance inspection shown in table III.

4.6 Examination.-

4.6.1 Equipment.- Each equipment shall be given an examination to determine that the design, material, and workmanship conform to this specification. The fit of parts shall be observed, with particular

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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reference to interchangeability of replaceable parts. Conformance to approved drawings with respect to material finish, construction, dimensions and weight, shall be verified. This examination should not include disassembly to the point that performance, durability, or appearance would be affected.

4.6.2 Repair parts.- All repair parts shall be subjected to an examination to ascertain that the materials, workmanship, and finish are first class in every respect and that they conform fully to the manufacturer's approved drawings. Resistance reading, at room temperature, shall be made upon all repair coils and resistors and the results checked against the design values of the original equipment. If the resistance, when corrected for difference in temperature, differs more than 10 percent from the design value, this shall be cause for rejection of the part. All coils and resistors shall be submitted to a dielectric test. The principal object of this inspection shall be to determine that the repair parts are equal to or better than the original parts. If the inspector has reason to doubt the ready interchangeability of the repair parts with the original parts, he may require a demonstration of such interchangeability.

4.6.3 Additional inspection.- Where other specifications form a part of this specification, examination and tests shall be conducted as required in the referenced specification.

4.7 Test procedures.-

4.7.1 Weight.-The weight of the equipment, and the weight of the repair parts, shall be taken and recorded. The weight shall not differ from the weight specified on the drawing of the equipment by more than 10 percent.

4.7.2 Creepage and clearance.- Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.7.3 Resistance.- The overall resistance of resistors and shunt coils shall be accurately measured to determine that they are within plus or minus 10 percent of the design values. The temperature at which the resistance readings are taken shall be recorded.

4.7.4 Current and power input.- The steady state current and power input with rated voltage at rated frequency applied shall be measured. The current shall be between plus and minus 10 percent of that shown on the master drawing. Measured values shall be recorded. Quality conformance tests shall consist of a check to show that the maximum steady state current values are not exceeded. Power input need not be measured during quality conformance tests.

4.7.5 General operation.- General operation shall include all the tests necessary to ascertain that the required operating features have been fully met. In particular, it shall be determined that the brake will set and release properly at normal voltage and frequency and in the normal position and that the brake can be released manually.

4.7.6 Inclination.- Equipment shall be tested for proper operation at rated voltage and frequency when inclined at 45 degrees in the forward, backward, left and right positions.

4.7.7 Voltage variation.- Brakes shall be tested to determine conformance with NEMA Publication ICI-11.44 A, B and C. Failure of the brake to release at 80 percent of rated voltage for d.c. or 85 percent of rated voltage for a.c. at rated frequency when cold (room ambient) shall be cause for rejection.

4.7.8 Temperature rise.- The test methods to be employed and the precautions to be observed in conducting temperature rise tests shall be in accordance with NEMA Publication ICI-2.41 and as specified in 4.7.8.1 through 4.7.8.5.

4.7.8.1 General requirement.- Temperature rise tests shall be made at rated voltage and frequency and the duty specified, to determine that the rises specified in 3.3.10 are not exceeded. Temperature rise tests shall be made on individual parts of the completely assembled equipment. The specified voltage shall be applied to all parts simultaneously so that the mutual heating effect of the several parts will be present.

4.7.8.2 Disc brakes.- Where both the brake and associated motor are manufactured in the same plant, the motor and brake shall be assembled and temperature rise tests on the brake made while the motor is running at rated conditions. Where the brake and associated motor are not manufactured at the

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same plant, it will be permissible to test the brake separately, provided the manufacturer demonstrates that in determining the temperature rise of the brake, adequate allowance is being made for heat transfer between the motor and brake. Adequate allowance would be provided by placing heating elements in close proximity to the motor end of the brake during test. The rating of the heating elements should be 20 percent of the horsepower of the largest motor with which the brake could be mated.

4.7.8.3 Starting temperature.- Temperature rise tests shall not be started if the ambient differs more than 5°C. from the ambient where the brake has stood during the preceding 2 hours.

4.7.8.4 Ambient variations.- A variation of more than 10°C. during a period of 6 hours, or a proportional change for runs of shorter duration, shall in no case be permitted. If the ambient variations are excessive or if the ambient changes rapidly at the end, the test shall be repeated.

4.7.8.5 Duration of tests.- The temperature rise test for continuously rated brakes shall be continued until the temperature rises that can be observed during the test have attained a steady final value. The temperature rise test for intermittently rated brakes shall be continued for the duty cycle specified for the brake.

4.7.9 Vibration.- Equipment shall be subjected to the vibration test specified in 3.3.12.2. Operation shall be checked at each frequency of vibration.

4.7.10 Insulation resistance.- Insulation resistance shall be as specified in 3.3.13. The measurement shall be made with an insulation-resistance-indicating meter with a full scale reading of 100 or 200 megohms, an open circuit voltage of 500 volts, and a voltage of at least 450 volts across a resistance of 1 megohm. The temperature and relative humidity shall be measured and recorded. Insulation resistance measurements shall be corrected to 25°C. Corrections shall be made on the basis of insulation resistance doubling for each 15°C. decrease in temperature. No correction shall be made for humidity.

4.7.11 Dielectric.- The dielectric test shall determine conformance to 3.3.14. The tests shall be made upon completely assembled equipment.

4.7.12 Shock.- Shock tests shall be conducted on all samples submitted for qualification tests in accordance with 3.3.12.3 with enclosure in place.

4.7.12.1 Method of mounting.- Equipment shall be mounted on the shock machine in a manner simulating normal shipboard mounting.

4.7.12.2 Definition of failure.-

(a) Breakage or appreciable distortion of any parts.
(b) Inability to perform principle functions.
(c) A value of insulation resistance lower than that specified in 3.3.13.
(d) Failure to pass a dielectric test of 65 percent of voltage specified in 3.3.14.
(e) Failure to pass inspection. The brake shall be disassembled following checks (a) through (d) and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the brake can be easily observed. Any observed effects of the shock should be recorded.

4.7.12.3 Disposal of shock-tested brakes.- Shock-tested brakes shall be disposed of as follows:

(a) Brakes which have been subjected to and successfully passed the HI shock test shall be considered acceptable provided post-shock tests are passed and corrective measures taken. Brakes shall be subjected to a temperature rise test in accordance with 4.7.8 and a dielectric test in accordance with 4.7.11. Minor deformations shall be corrected. Minor deformations are defined as those which do not cause rejection, but which are in excess of dimensional tolerances specified on the brake drawing.
(b) Brakes which have been subjected to the shock test and have failed structurally will not be acceptable. Brakes which have failed due to fastenings, fittings, or other nonstructural parts may be reconditioned in accordance with item (a) and resubmitted for shock tests.

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4.7.13 Torque.- Brakes shall be subjected to a static torque test to determine that the maximum torque delivered by the brake equals or exceeds that shown on the brake drawing.

4.7.14 Enclosures.-

4.7.14.1 Qualification tests.- Qualification tests are required only on explosionproof enclosures and those enclosures included with complete equipments submitted for qualification approval. Explosionproof enclosures shall be tested by the Bureau of Mines.

4.7.14.2 Comparison inspection.- Comparison inspection shall be conducted on a representative sample of each group of similarly constructed enclosures at intervals of 3 years. All enclosures in a group shall be similarly constructed, and shall have the same features as to doors, louvers, and degree of enclosure. The volume of the largest enclosure in the group shall not exceed four times the volume of the smallest.

4.7.14.3 Effectiveness of enclosure.- Enclosures shall be inspected to determine compliance with 3.4.1 and as specified in 4.7.14.3.1, 4.7.14.3.2 and 4.7.14.3.3.

4.7.14.3.1 Dripproof.- No tests are required. An examination shall be made to determine that falling moisture or dirt will not interfere with the mechanical or electrical operation of the equipment enclosed when inclined to any angle not exceeding 45 degrees from the vertical.

4.7.14.3.2 Watertight.- At the option of the manufacturer, the test may consist of submerging the equipment to a depth of 3 feet for 5 minutes in lieu of the NEMA hose test. Equivalent pressure or vacuum tests, with the enclosure submerged, are acceptable provided that the resulting pressure is on the outside of the enclosure.

4.7.14.3.3 Gaskets and seals.- Before conducting tests, all gasketed joints and seals (cable entrance plates excluded) shall be broken and reassembled three times.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Brakes.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be packaged in heat sealable, laminated paper and foil bags or heat sealable plastic film. Repair parts shall be identified by Federal stock number.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation and packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Torque (see 3.3.2).
(c) Voltage (see 3.3.3).
(d) Duty (see 3.3.5).
(e) Mounting (see 3.3.7).
(f) Type of enclosure (see 3.4).
(g) Number of repair parts sets required (see 3.5.2).
(h) Preservation and packaging, packing, and marking requirements if other than as specified in 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 16392, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C. and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6110-N001(NAVY))

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

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ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO — IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification.) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

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